

LINEAR REGRESSION

Linear Regression is of two the Supervised Machine Learning model in which the model finds the best fit linear line between the independent and dependent variable i.e. it finds the linear relationship between the dependent and independent variable.

Linear Regression is of two types i.e.

- (1) Simple Linear Regression
- (2) Multiple Linear Regression

* Simple Linear Regression is where only one independent variable is present and the model has to find the linear relationship of it with the dependent variable.

* In Multiple Linear Regression there are more than one independent variables for model to find the relationship.

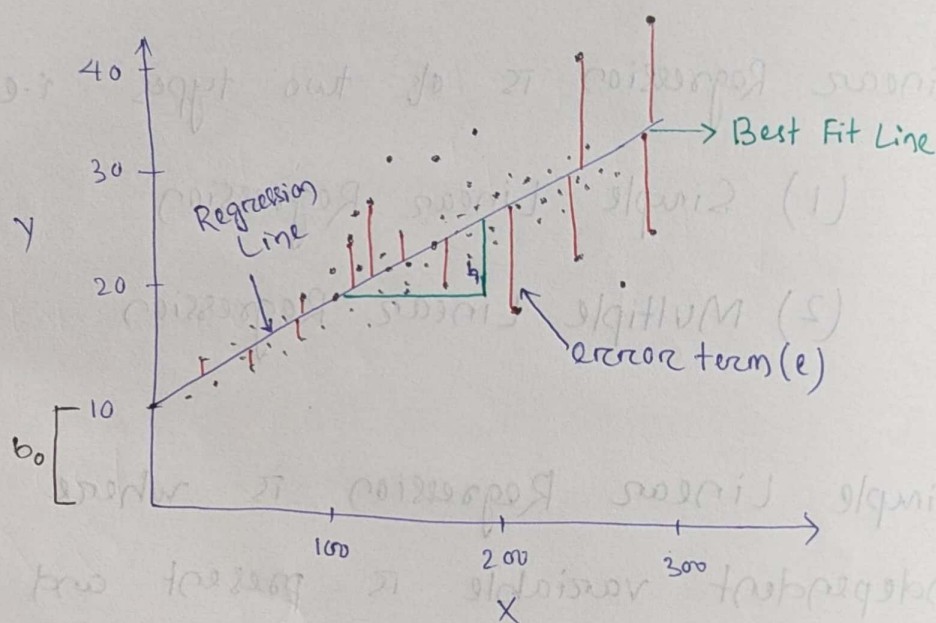
$$y = mx + c$$

↳ Linear Regression Equation.

Equation of Simple LR

$$y = b_0 + b_1 x$$

- * Here b_0 is the intercept, b_1 is coefficient or slope, x is the independent variable and y is the dependent variable.



- * Black dots are the data points i.e. the actual values.

→ b_0 is the intercept which is 10 and b_1 is the slope of the x -variable.

→ The blue line is the best fit line predicted by the model i.e. the predicted values lie on the blue line.

The Vertical distance (red line) between the data point and the regression line is known as error or residual.

Each data point has one residual and the sum of all the difference is known as the Sum of Residuals / Error.

We will calculate the Residual Sum of Square for each line and which ever line will give less Residual Sum of Square, that will be selected as the Best Fit Line for our Regression model.

$$\text{Residual/Error} = \text{Actual} - \text{Predicted values}$$

Residual Sum of Square

$$= \left(\sum (\text{Actual} - \text{Predicted}) \right)^2$$

i.e

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\hat{y}_i = mx_i + c$$

Note:-

The Line which is minimum Residual Sum of Square Error that will be choose as the Best Fit line.

* ~~Regression~~

Above Equation is also known as cost functions.

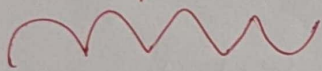
* Linear Regression is used when the dependent variable is of continuous data type and independent variables could be of any data type like continuous, nominal/categorical etc.

Equation of Multiple LR :-

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 \dots + b_nx_n$$

Here b_0 is the intercept, $b_1, b_2, b_3 \dots b_n$ are co-efficients or slopes of the independent variables $x_1, x_2, x_3 \dots x_n$ and y is the dependent variable.

R squared :-



R squared is the most commonly used metric for model evaluation in regression analysis. The value of R squared lies between 0 to 1, the value closer to 1 the better the model.

Basically R^2 is how much variation my model is able to understand.

* R^2 is the model Accuracy in Linear Regression there is no confusion matrix here.

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}}$$

$$= 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Here,

SS_{RES} = Residual Sum of Squares

SS_{TOT} = Total Sum of Squares

$\bar{y}$ = median mean of the sample

y_i = the value in a sample

$\hat{y}_i$ = the value estimated by the regression line.

Total Sum of Squares :-

The total sum of squares quantifies the total variation in a sample.

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2$$

Sum of Squares :-

The regression sum of squares describes how well a regression model represents the model data.

A higher regression sum of squares indicates that the model does not fit the data well.

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

~~Resid~~

Residual Sum of square :-

The Residual Sum of Squares essentially measures the variation of modeling error.

Generally, a lower residual sum of squares indicates that the regression model can better explain the data while a higher higher residual sum of squares indicates that the model poorly explains the data.

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

The relationship between the three types of sum of square can be summarized by the following equation:

$$TSS = SSR + SSE$$

Adjusted R squared :-

Adj- R^2 indicates the that Per unit change in R^2 values value when we add or remove any features.

It is the improvement to R squared. The problem / drawback with R^2 is that as the features increase, the value of R^2 also increases which gives the illusion of a good model, but we have no idea which are the specific features which helps to increase the R^2 value.

So the Adjusted R^2 solves the drawback of R^2 . It only considers the features which are important for the model and shows the real improvement of the model.

Note :-

Adj- R^2 is always lower than R^2 .

$$R^2 - \text{Adjusted} = 1 - \frac{(1 - R^2)(N - 1)}{N - K - 1}$$

Here,

R^2 = sample R^2 squared

K = Number of independent (x) variables

N = Total sample size

Example

Ex 1

Exp

Sol

$$R^2 = 0.9$$

$$\text{Adj-}R^2 = 0.8$$

MODEL - 1

2	20L
3	30L

→ This two data our model is not able to understand hence in gave 90% accuracy instead of 100%

Ex 2:-

Exp	Sal	Cabs	
2	2L	1	$R^2 = 0.94$
2	2L	1	$Adj-R^2 = 0.8$
2	2L	1	
2	2L	1	
2	2L	1	
3	3L	1	
3	3L	1	
3	3L	1	
2	2OL	1	
3	3OL	1	

MODEL - 2

Ex 3:-

Exp	Sal	Skills	Cabs	
2	2L	1	1	$R^2 = 0.95$
2	2L	1	1	$Adj-R^2 = 0.84$
2	2L	1	1	
3	3L	1	1	
3	3L	1	1	
3	3L	1	1	
2	2OL	ML	1	
3	3OL	ML	1	

* Cabs is a zero variance features in Ex 2 & Ex 3.

Python

Adjusted R^2 represented how much of your R^2 value increased or decreased, when per unit changes of your R^2 into your R^2 value happen when you are added or remove a specific features.

It can give you the importance of a variable in your model.

- ⇒ In the above example we have created a model with two variable i.e. Exp & Sal and it gave some R^2 and Adj- R^2 value.
- ⇒ In 2nd model we have add one more variable called Cabs but still our model giving us the same R^2 and Adj- R^2 values that means this newly added variable is not giving any importance in our model. We can drop that features instead of using.
- ⇒ In model-3 when we add skills features then our R^2 and Adj- R^2 value got increase that means this value is important in our model for increasing the model Accuracy.

Below are the different ways to which will help us to improve R^2 value,

- Cross Validation
- Zero Variance check
- Outlier detection
- Standardization or Normalization
- Correlation check

Standardization :-

Standardization is a technique which help us to do feature scaling in ML, ~~also~~ where mean is 0 and standard deviation is 1.

$$Z = \frac{x - \mu}{\sigma}$$

Feature Scaling :-

The goal of applying Feature Scaling is to make sure features are on almost the same scale so that each feature is equally important and make it easier to process by most ML algorithms.

Q. When to standardize data and Why?

Ans.

Ex 1:-

<u>Sqft</u>	<u>BedRoom</u>	<u>Bathroom</u>	<u>Price</u>
4000	4	2	40L
3000	3	2	30L
2000	2	1	20L
3000	7	2	30L
4000	4	2	40L

On the above example we need to perform standardization otherwise when our Linear Regression Model will calculate the internally do the calculation for Price prediction then Sqft features will give more weightage and other two features value will not at all give any importance at the time of calculation due to the huge difference in range.

So to give the important +

When it will predict the price, then it will not give any important to Bedroom & Bathroom features and because of that it will predict the wrong price. Above is the example where Sqft = 3000, Bedroom = 7 but still it is predicting the price as 30 L.

To avoid these kind of problem we need to standardize our data so that it will give equal importance to all the features for model prediction, and it will not predict the model based on one or two feature which are more weightage.

Below are the algorithm

```
from sklearn.preprocessing import StandardScaler
```

```
scaling = StandardScaler()
```

```
scaling.fit_transform(ds[['Sqrft', 'BedRoom', 'Bathroom']])
```

* StandardScaler will help you to standardize our data.

Normalization

Normalization is also a technique which help us to do Feature Scaling in Machine Learning where it rescale the real-value into a 0 to 1 range. It is also known as Min-Max scaling.

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Here X_{max} and X_{min} are the maximum and the minimum values of the features respectively.

from sklearn.preprocessing import MinMaxScaler

scaling = MinMaxScaler()

scaling.fit_transform(df[['Sqrft', 'BedRoom', 'Bathroom']])

Note :-

* Standardization does not affect much by outlier but in case of Normalization it's badly affected by the outlier, because it uses maximum and minimum value to normalize the data.

Hence before using Normalization we must have to perform outlier detection.

Q:- When to use which technique to scale down the value?

Ans. :-

- Normalization is good to use when you know that the distribution of your data does not follow a Gaussian distribution.

This can be used & useful in algorithms that do not assume any distribution of

the data like K-Nearest Neighbors and Neural Networks.

- standardization, on the other hand can be used or helpful in cases where the data follows a Gaussian distribution. Also However, this does not have to be necessarily true. Also, unlike normalization, standardization does not have a bounding range.

So, even if you have outliers in your data, they will not be affected by standardization.

However at the end of the day, the choice of using Normalization or standardization will depend on your problem and ML algorithms, you are using. There is no hard and fast rule to tell you when to normalize or standardize your data.

Linear Regression Code :-



```
from sklearn import linear_model
```

```
reg = linear_model.LinearRegression()
```

```
reg.fit(X_train, y_train)
```

from sklearn.model_selection import

```
from sklearn.linear_model import LinearRegression
```

```
reg = LinearRegression()
```

```
reg.fit(X_train, y_train)
```

Q. How to check Accuracy Score?

```
from sklearn.metrics import r2_score
```

```
y_pred = reg.predict(X_test)
```

```
AccScore = r2_score(y_test, y_pred)
```

```
reg.fit().summary()
```

↳ This will give the summary about our model & like,

→ Dependant variable name

→ Which model it use (Ex:- OLS, Linear Regression, Logistic etc).

→ R-squared value

→ Adj R² value

→ method etc.